Component Showcase

Living documentation for homework-template

COURSE 101
Course Name

Student Student Name
Instructor Instructor Name
Collaborators None

Student ID Student ID
Due date Not applicable

1 Problems and solutions

Problem 1: A numbered problem

12 points

Problem statements may contain prose, displayed mathematics, lists, and references. The optional argument records the available points.

(a) The first subproblem.

(b) The second subproblem.

Solution

Solutions use a distinct breakable panel and may span multiple pages.

Final answer

Use an answer box when the final result should be easy to locate.

Problem 2: A problem without points

The points argument is optional, but the descriptive title is required.

2 Mathematics

Theorem 1 (Geometric series)

For $r \neq 1$ and every integer $n \geq 0$,

$$\sum_{k=0}^{n} r^k = \frac{1 - r^{n+1}}{1 - r}.$$

Proof. Multiply the sum by $1 - r$ and cancel adjacent terms. □

The template includes theorem, lemma, proposition, corollary, definition, and remark environments. Equations, theorems, figures, and tables can be referenced with `\cref`.

3 Images and diagrams

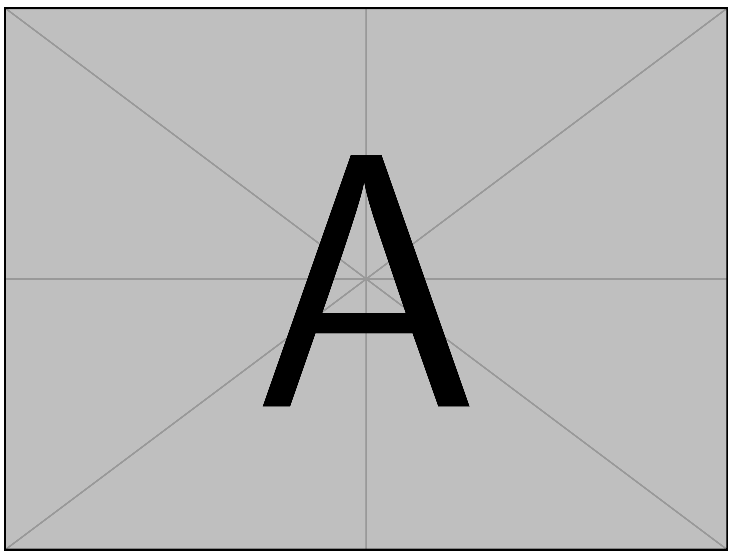

Figure 1: A sample raster-compatible figure included by the TeX distribution.

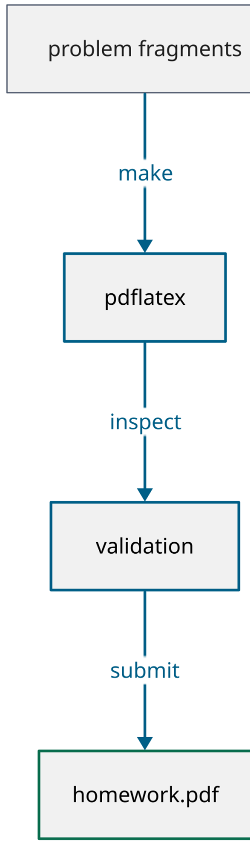

Figure 2: The source-to-submission workflow used by the template.

Figures such as figure 1 accept an explicit visual description. Generated diagram PDFs remain committed, so changing a diagram requires D2 but compiling a document does not.

4 Code, tables, and citations

Binary search

```
def binary_search(values, target):
    low, high = 0, len(values) - 1
    while low <= high:
        middle = (low + high) // 2
        if values[middle] == target:
            return middle
        if values[middle] < target:
            low = middle + 1
        else:
            high = middle - 1
    return None
```

Table 1: Asymptotic costs of common search operations.

Method	**Prerequisite**	**Worst case**
Linear search	None	$\Theta(n)$
Binary search	Sorted input	$\Theta(\log n)$
Hash lookup	Suitable hash table	$\Theta(n)$

Table 1 uses spacing rather than decorative rules or vertical separators. Bibliographic citations use BibTeX and Natbib; for example, see Knuth (1984).

References

Donald E. Knuth. Literate programming. *The Computer Journal*, 27(2):97–111, 1984. doi: 10.1093/comjnl/27.2.97.